Supply chain Optimization

Solutions Plus is an industrial chemicals company that produces specialized cleaning fluids and solvents for a wide variety of applications. Solutions Plus has received an invitation to submit a bid to supply Great North American Railroad with a cleaning fluid for locomotives. Great North American Railroad needs the cleaning fluid at 11 locations (railway stations), and it has provided the following information to Solutions Plus regarding the number of gallons of cleaning fluid required at each location (see the following table).

Solutions Plus can produce the cleaning fluid at its Cincinnati plant for $1.20 per gallon. Even though the Cincinnati location is its only plant, Solutions Plus has negotiated with an industrial chemicals company located in Oakland, California, to produce and ship up to 500,000 gallons of the locomotive cleaning fluid to selected Solutions Plus customer locations.

The Oakland company will charge Solutions Plus $1.65 per gallon to produce the cleaning fluid, but Solutions Plus thinks that the lower shipping costs from Oakland to some customer locations may offset the added cost to produce the product.

Location	Gallons Required	Location	Gallons Required
Santa Ana	22,418	Glendale	33,689
El Paso	6,800	Jacksonville	68,486
Pendleton	80,290	Little Rock	148,586
Houston	100,447	Bridgeport	111,475
Kansas City	24,570	Sacramento	112,000
Los Angeles	64,761		

Table: Gallons of Cleaning Fluid Required at Each Location

The president of Solutions Plus, Chanda Weaver, contacted several trucking companies to negotiate shipping rates between the two production facilities (Cincinnati and Oakland) and the locations where the railroad locomotives are cleaned. The following table shows the quotes received in terms of dollars per gallon. The "—" entries in the table identify shipping routes that

will not be considered because of the large distances involved. These quotes for shipping rates are guaranteed for one year.

Location	Cincinnati	Oakland
Santa Ana	—	0.22
El Paso	0.84	0.74
Pendleton	0.83	0.49
Houston	0.45	—
Kansas City	0.36	—
Los Angeles	—	0.22
Glendale	—	0.22
Jacksonville	0.34	—
Little Rock	0.34	—
Bridgeport	0.34	—
Sacramento	—	0.15

Table: Freight Cost ($ per gallon)

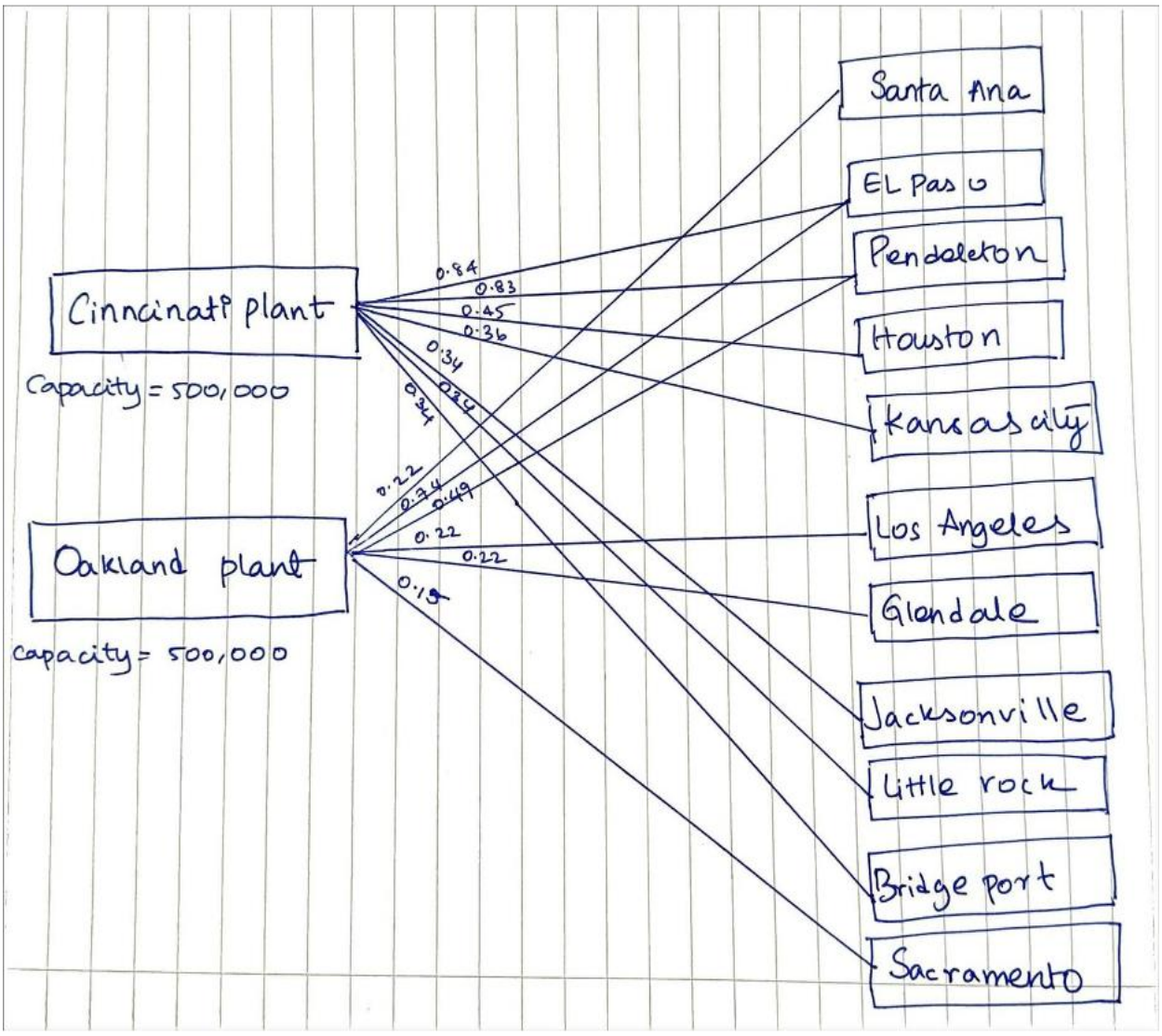
To submit a bid to the railroad company, Solutions Plus must determine the price per gallon it will charge. Solutions Plus usually sells its cleaning fluids for 15% more than the cost to produce and deliver the product. For this big contract, however, Fred Roedel, the director of marketing, suggested that maybe the company should consider a smaller profit margin. In addition, to ensure that Solutions Plus will have adequate capacity to satisfy existing orders as well as accept orders for other new business if it wins the bid, the management team decided to limit the number of gallons of the locomotive cleaning fluid produced in the Cincinnati plant to at most 500,000 gallons.

Problem Statement:

Solution Plus is a chemical company aims to optimise its production and distribution strategy for a cleaning fluid contract from Great north American railroad. The company aims to find how much quantity should be shipped from each facility to each location. We will use transportation optimization method to determine optimal allocation of production and distribution at a minimized cost.

(a) **If Solutions Plus wins the bid, which production facility (Cincinnati or Oakland) should supply the cleaning fluid to the locations where the railroad locomotives are cleaned? How much should be shipped from each facility to each location?**

Decision variables



As shown in the above diagram we have 13 feasible routes from the production facility to railroad locomotive cleaning locations.

x_{ij} - The number of gallons of cleaning fluid shipped from facility i to customer location j.

where $i = \{c, o\}$; c: cinninati ; o: Oakland.

Objective function

The aim is to reduce the production and transportation cost while satisfying the demands at all the location.

$$z = \sum_{j=1}^{11} (1.20 \cdot x_{cj} + f_{cj} \cdot x_{cj}) + \sum_{j=1}^{11} (1.65 \cdot x_{oj} + f_{oj} \cdot x_{oj})$$

where,

f_{cj} : freight cost from Cinncinati to location j

f_{oj} : freight cost from Oakland to location j.

x_{cj} = number of cleaning fluid from Cincinnati facility to location j

x_{oj} = number of cleaning fluid from Oakland facility to location j

Production capacity constraint

Cinncinati production limit = 500,000

$$\sum_{j=1}^{11} x_{cj} \leq 500{,}000$$

Oakland production limit = 500,000

$$\sum_{j=1}^{11} x_{oj} \leq 500{,}000$$

Demand constraint for each location and restricted shipping routes

$x_{cj} + x_{oj} = d_j$ for all $j = \{1, 2, 3, \ldots\ldots, 10, 11\}$

d_j is the demand at location j.

Restricted shipping routes:-

$x_{cj} = 0$ for locations where cincinnati shipping is unavailable.

$x_{oj} = 0$ for locations where Oakland shipping is unavailable.

Total mathematical representation

```
Minimize Z = 1.2*xc + 1.65*xo + 0.22*xso + 0.74*xeo + 0.49*xpo + 0.22*xlo + 0.22*xgo + 0.15*xsa
                   + 0.84*xec + 0.83*xpc + 0.45*xhc + 0.36*xkc + 0.34*xjc + 0.34*xlc + 0.34*xbc
Subject to

supply constraints
xso + xeo + xpo + xlo + xgo + xsa <= 500000       # Supply constraint for Oakland facility
xec + xpc + xhc + xkc + xjc + xlc + xbc <= 500000 # Supply constraint for Cincinnati facility

Demand constraints
xso == 22418           # Demand constraint for Santa Ana
xec + xeo == 6800      # Demand constraint for El paso
xpc + xpo == 80290     # Demand constraint for Pendelton
xhc == 100447          # Demand constraint for Houston
xkc == 24570           # Demand constraint for Kansas city
xlo == 64761           # Demand constraint for los angeles
xgo == 33689           # Demand constraint for Glendale
xjc == 68486           # Demand constraint for Jacksonville
xlc == 148586          # Demand constraint for little rock
xbc == 111475          # Demand constraint for bridgeport
xsa == 112000          # Demand constraint for Sacramento
```

Solution:

Demand Location	Demand (Gallons)	From Cincinnati (Gallons)	From Oakland (Gallons)
Santa Ana	22,418	0	22,418
El Paso	6,800	6,800	0
Pendleton	80,290	39,636	40,654
Houston	1,00,447	1,00,447	0
Kansas City	24,570	24,570	0
Los Angeles	64,761	0	64,761
Glendale	33,689	0	33,689
Jacksonville	68,486	68,486	0
Little Rock	1,48,586	1,48,586	0
Bridgeport	1,11,475	1,11,475	0
Sacramento	1,12,000	0	1,12,000

The total minimum cost including production and transportation cost is **$1,318,984.93.** The total demand combining all the location is **773,522** gallons, the Cincinnati facility should produce and send 500,000 gallon and the remaining **273,522** gallon is produced by the Oakland facility.

(b) **What is the breakeven point for Solutions Plus? That is, what is the lowest bid Solutions Plus can submit without losing money?**

The lowest amount Solution Plus can bid is **$1,318,984.93** without losing any money.

(c) **If Solutions Plus wants to use its standard 15% markup, how much should it bid?**

If Solution plus uses the standard mark Up rate, then the bid price will be

Bid Price = Total Cost × (1 + 0.15)

Bid price = Bid price = 1,318,984.93 × 1.15 = $**1,516,832.67**

(d) **Freight costs are significantly affected by the price of fuel. The contract on which Solutions Plus is bidding is for two years. Discuss how fluctuation in freight costs might affect the bid Solutions Plus submits.**

The contract solution plus is bidding for 2 years, but the transportation costs are guaranteed only for a year. Considering the increase in freight costs for the second year, Solution Plus can increase its mark from **15%** to **20%**, i.e., **$ 1582781.916**. This will absorb the fluctuation in freight cost and save from a loss, but there is a high chance of losing the bid. Bidding **a 15%** mark-up price ensures that they may win the bid; it also provides both profitability and allows Solution Plus to cover the fluctuations in freight costs.

(e) **How would changes in the cost of production or shipping rates impact the overall cost and the optimal shipping strategy? Analyse scenarios where the costs increase or decrease by 10%.**

10% increase in production and transportation cost will increase the overall cost to **$1,450,883.423**, while the overall cost **is $1,187,086.437** if the production and shipping cost are decreased by 10%. The quantity shipped from Cincinnati and Oakland facility remains the same, the optimal solution depends heavily on the production cost and capacity that the transportation cost.